$$I_{RMS} = \sqrt{\frac{I_m^2\,\pi}{2\pi}} = \frac{I_m}{\sqrt{2}}$$

$$I_{rms} = 0.707\, I_m \qquad (4.35)$$

Thus we find that for a symmetrical sinusoidal current rms value of current is 70.7 % of its peak value.

Similarly for alternating voltage, it can be shown that

$$V_{rms} = 0.707\, V_m \qquad (4.36)$$

Note

RMS value of alternating current is also called effective value and is represented as I_{eff}. It is used to compare RMS current of AC to an equivalent steady current.

RMS value is also defined as that value of the steady current which when flowing through a given circuit for a given time produces the same amount of heat as produced by the alternating current when flowing through the same circuit for the same time. The effective value of an alternating voltage is represented by V_{eff}.

Note

For example, if we consider n currents in one cycle of AC, namely i_1, i_2, ... i_n, then RMS value is given by

$$I_{RMS} = \sqrt{\frac{\text{Sum of squares of all currents over one cycle}}{\text{Number of currents}}}$$

$$I_{RMS} = \sqrt{\frac{i_1^2 + i_2^2 + \ldots + i_n^2}{n}}$$

DO YOU KNOW?

For common household appliances, the voltage rating and current rating are generally specified in terms of their RMS value. The domestic AC supply is 230V, 50 Hz. It is the RMS or effective value. Its peak value will be $V_m = \sqrt{2}\ V_{rms} = \sqrt{2} \times 230 = 325\,\text{V}$.

EXAMPLE 4.18

Write down the equation for a sinusoidal voltage of 50 Hz and its peak value is 20 V. Draw the corresponding voltage versus time graph.

Solution

$f = 50\,\text{Hz};\ V_m = 20\ \text{V}$

Instantaneous voltage, $v = V_m \sin\omega t$

$= V_m \sin 2\pi f t$

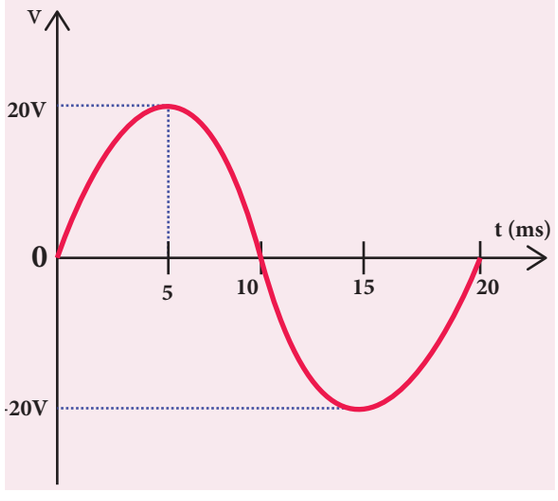
$= 20\sin(2\pi \times 50)\,\text{t} \ = 20\sin(100 \times 3.14)\,\text{t}$

$v = 20\sin 314\,\text{t}$

Time for one cycle, $T = \frac{1}{f} = \frac{1}{50} = 0.02\,\text{s}$

$= 20 \times 10^{-3}\,\text{s} = 20\,\text{ms}$

The wave form is given below.



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